DDA5001 MACHINE LEARNING FINAL PROJECT - PART II

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ABSTRACT

This report presents the second part of the final project by Group 10 for DDA5001 Machine Learning, Fall 2025. In this study, we fine-tuned the Qwen3-0.6B-Base model using three distinct optimizers: SGD, Adam, and LoRA, each under various hyperparameter configurations. Training and validation losses were analyzed over up to three epochs on the Math500 dataset. Our findings indicate that while all optimizers achieved some degree of convergence, the Adam optimizer provided the lowest overall validation loss with the most stable performance. Meanwhile, the SGD optimizer, when configured with 2 epochs and a learning rate of 1e-4, achieved the highest accuracy (14.40%) on the Math500 test set among the fine-tuned models. Resource usage comparisons revealed that LoRA was the most memory-efficient optimizer, whereas Adam consumed more GPU memory but offered faster training times per epoch. These results highlight the importance of careful hyperparameter tuning and the trade-offs between computational efficiency and model performance in deep learning tasks.

1 Analysis of Training and Validation Loss Curves

1.1 LoRA Optimizer

Figure 1 displays the training loss of LoRA models with ranks 4, 8, and 16 (lr = 2e-5). All converge quickly within 200 steps, but rank=16 achieves the lowest and smoothest loss, while rank=4 shows noticeable oscillations—especially between steps 200–600—suggesting limited capacity. Rank=8 offers a balanced trade-off. Higher ranks thus improve stability and performance at the cost of greater resource use.

1.2 Adam Optimizer

Figure 2 displays the training loss of Adam models with learning rates 1e-5 and 2e-5. Both configurations show rapid initial convergence, but significant fluctuations persist, especially at later steps. The lr=2e-5 setting achieves lower loss earlier, while the lr=1e-5 variant exhibits a sharp dip near step 2000, possibly due to instability.

[1] Project coordination and report structure design.
[2] LoRA experiment execution and analysis.
[3] Adam experiment execution and analysis.
[4] SGD experiment execution and analysis.
[5] Validation performance and comprehensive comparison.
[6] Resource and efficiency evaluation.
[7] Model performance assessment and test results.
[8] Challenges summary, observations synthesis, and abstract writing.

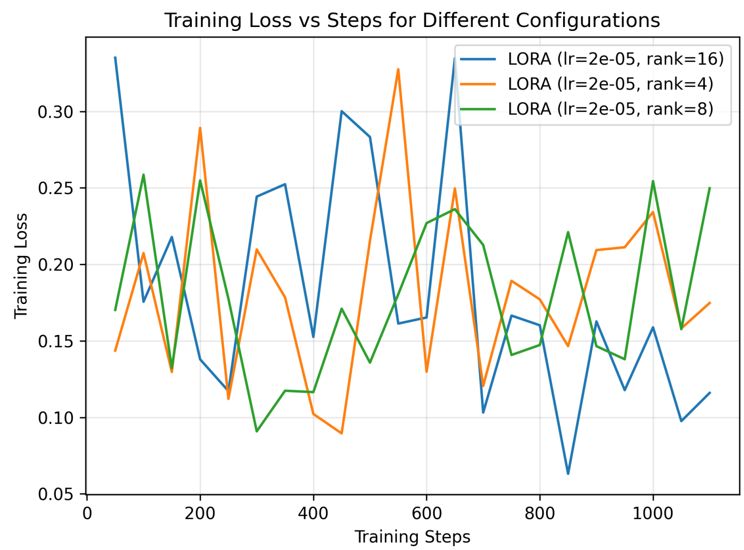

Figure 1: Training loss of LoRA models with ranks 4, 8, and 16, trained at a learning rate of $2e^{-5}$.

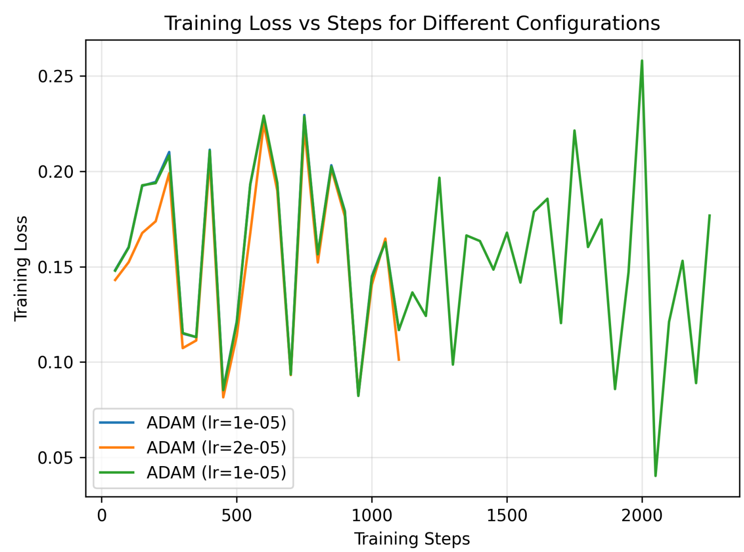

Figure 2: Training loss of Adam models with learning rates $1e^{-5}$ and $2e^{-5}$.

1.3 SGD Optimizer

Figure 3 displays the training loss of SGD models with learning rates 0.0001 and 0.001. Both show large fluctuations and unstable convergence, with lr=0.001 descending faster initially but exhibiting more extreme oscillations. The results highlight SGD's sensitivity to learning rate and its tendency toward noisy training dynamics.

1.4 Validation Loss Comparison

Figure 4 displays the validation loss across different optimizer configurations over training steps. The Adam model with lr=2e-5 (orange) achieves the lowest validation loss, showing consistent and smooth convergence, indicating strong generalization performance. The LoRA models (red, purple, brown) exhibit steady declines but plateau earlier than Adam, with rank=16 (red) performing slightly better than lower ranks. SGD models (pink, gray, yellow) show higher initial validation loss and slower improvement; in particular, the SGD (lr=0.001) variant (yellow) remains the highest

Figure 3: Training loss of SGD models with learning rates 0.0001 and 0.001.

throughout, suggesting poor generalization. Overall, Adam demonstrates superior stability and generalization, while LoRA offers moderate performance with less computational cost.

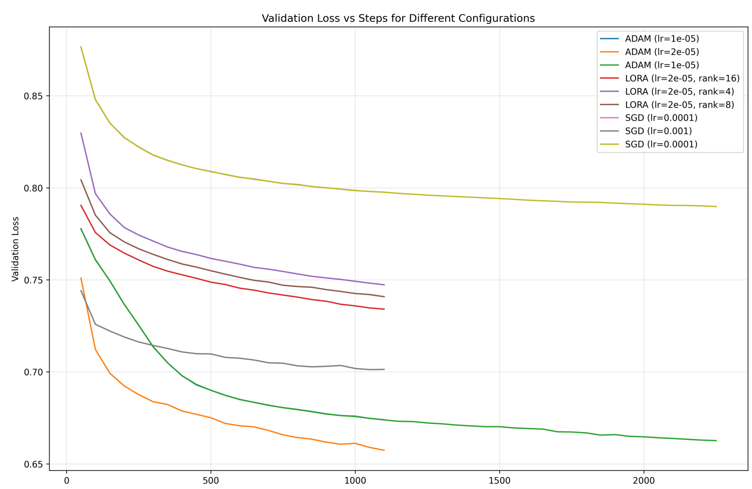

Figure 4: Validation loss by different optimizers with different hyperparameters.

2 Resource Usage Comparison

Table 1 presents a comparison of resource usage across LoRA, SGD, and Adam optimizers. LoRA consistently requires the least memory (4.85–4.97 GB), making it the most memory-efficient option, while Adam consumes the highest peak memory (7.92 GB) due to its adaptive momentum computation. SGD lies in between with 5.68 GB usage regardless of learning rate or epoch count. In terms of training time, Adam is fastest per epoch (around 816–828 seconds), followed by SGD (799–821 s), and LoRA (1121–1126 s). Notably, doubling the number of epochs roughly doubles training time for all methods, but Adam remains faster than LoRA despite higher memory cost. These results highlight a trade-off: LoRA offers efficiency at lower memory, while Adam provides faster convergence at the expense of higher GPU utilization.

Table 1: Resource usage comparison across optimizers.

Optimizer	Optimizer Detail	Max Memory Allocated	Training Time (s)
LoRA	num_epochs=1, lr=2e-5, rank=8	4.89 GB	1125.71
	num_epochs=1, lr=2e-5, rank=16	4.97 GB	1121.43
	num_epochs=1, lr=2e-5, rank=4	4.85 GB	1125.74
SGD	num_epochs=1, lr=1e-3	5.68 GB	798.81
	num_epochs=1, lr=1e-4	5.68 GB	821.06
	num_epochs=2, lr=1e-4	5.68 GB	1610.66
Adam	num_epochs=1, lr=2e-5	7.92 GB	816.83
	num_epochs=1, lr=1e-5	7.92 GB	827.65
	num_epochs=2, lr=5e-6	7.92 GB	1627.69

3 Model Performance on Math500 Test Set

Table 2 presents the accuracy of various fine-tuned models on the Math500 test set. The baseline model without fine-tuning achieves 10.00% accuracy, serving as a reference point. Among all configurations, SGD with 2 epochs and learning rate 1e-4 achieves the highest performance at 14.40%, significantly outperforming other settings. LoRA models show moderate improvements over the baseline, with the rank=4 variant reaching 13.00%, suggesting that lower ranks may capture useful patterns efficiently. Adam performs well with lr=1e-5 (13.20%), but increasing epochs does not improve results, indicating potential saturation. Overall, SGD demonstrates the strongest generalization capability under extended training, while LoRA offers competitive performance with reduced parameter updates.

Table 2: Model accuracy on the Math500 test set.

Model	Optimizer Detail	Accuracy (%)
Baseline (no fine-tuning)	—	10.00
LoRA	num_epochs=1, lr=2e-5, rank=8	12.80
	num_epochs=1, lr=2e-5, rank=16	12.00
	num_epochs=1, lr=2e-5, rank=4	13.00
SGD	num_epochs=1, lr=1e-3	10.80
	num_epochs=1, lr=1e-4	13.20
	num_epochs=2, lr=1e-4	14.40
Adam	num_epochs=1, lr=2e-5	12.60
	num_epochs=1, lr=1e-5	13.20
	num_epochs=2, lr=5e-6	12.60

4 Challenges and Observations

4.1 Key Challenges

- Hyperparameter sensitivity varied significantly across optimizers, requiring extensive trial-and-error to identify effective configurations.
- Balancing computational efficiency (memory/time) with model performance posed a non-trivial trade-off, especially under hardware constraints.

4.2 Observations

- LoRA provided the most memory-efficient training, but required careful tuning of the rank parameter to ensure stable convergence.
- Adam exhibited stable and predictable training dynamics, though at the cost of significantly higher GPU memory usage.
- SGD was highly sensitive to the learning rate; however, its performance improved substantially with extended training (e.g., 2 epochs), achieving the highest test accuracy on Math500.
- All optimizers showed a consistent gap between training and validation loss, indicating mild overfitting—suggesting that regularization or early stopping could further improve generalization.